

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)



Forensic Examination Report

SBT-DF204 Case Study 2: Chrome History Forensics & Cryptocurrency Analysis

Case Name:	SBT-DF204 Case Study 2: Chrome History Forensics & Cryptocurrency Analysis
Analyst:	Abubakari-Sadique Hamidu
Date of Report:	October 10, 2026

Evidence Source:	Google Chrome SQLite Database (History)
SHA-256 Hash:	B991b0fa69662c44caf599e468c8088576eb d9cebbf46d02c89cf51bcde3fd19

1. Executive Summary

This forensic investigation was conducted to analyze browser activity extracted from a suspect Google Chrome web browser database (History). The principal objectives were to reconstruct the user's chronological browsing session, determine user intent during online activities, investigate the creation of illicit classified advertisements, trace email communications, identify cryptocurrency transaction verifications, and cross-reference downloaded digital artifacts.

The forensic analysis established that on April 19, 2022, the primary suspect (profile FSCS_User) conducted targeted web searches leading to Craigslist Baltimore, created and managed a live listing titled "cheaper than Rx supplements" (Ad ID: 7473121658), engaged in related email communications via unsub.fscs@gmail.com, queried specific Bitcoin transaction records on Blockchain Explorer, and downloaded a proof-of-payment image file (proof_of_payment.png) originating from Imgur.

2. Evidence Integrity & Environment Verification

2.1 Evidence Chain of Custody & Hash Verification

To preserve forensic integrity and ensure strict adherence to evidence preservation standards, all analysis was performed on a bit-stream duplicate copy of the primary evidence file (History).

The primary file was set to read-only status (chmod 444).

Field	Value
Original Evidence File Path	evidence/History
Working Copy Path	evidence/History_working_copy.db
File Size	196,608 bytes
Cryptographic Checksum (SHA-256)	b991b0fa69662c44caf599e468c8088576ebd 9cebbf46d02c89cf51bcde3fd19

```

sadique@Abubakari-Sadiques-MacBook-Pro SBT-DF204-Lab2 % ls -la /Users/sadique/Desktop/Assignments-ICDFA/forensics/SBT-DF204-Lab2
total 0
drwxr-xr-x  4 sadique  staff  128 Oct  9 22:25 .
drwxr-xr-x 18 sadique  staff  576 Oct  9 22:25 ..
drwxr-xr-x  2 sadique  staff   64 Oct  9 22:25 evidence
drwxr-xr-x  2 sadique  staff   64 Oct  9 22:25 screenshots
sadique@Abubakari-Sadiques-MacBook-Pro SBT-DF204-Lab2 % cp ~/Downloads/case_study_2/History /Users/sadique/Desktop/Assignments-ICDFA/forensics/SBT-DF204-Lab2/evidence/History
sadique@Abubakari-Sadiques-MacBook-Pro SBT-DF204-Lab2 % ls -la /Users/sadique/Desktop/Assignments-ICDFA/forensics/SBT-DF204-Lab2/evidence/
total 384
drwxr-xr-x  3 sadique  staff   96 Oct 10 07:45 .
drwxr-xr-x  4 sadique  staff  128 Oct  9 22:25 ..
-rw-r--r--@ 1 sadique  staff 196608 Oct 10 07:45 History
sadique@Abubakari-Sadiques-MacBook-Pro SBT-DF204-Lab2 %

```

```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF204-Lab2 — docker exec -it sbtdf203-lab /bin/bash — 156x36

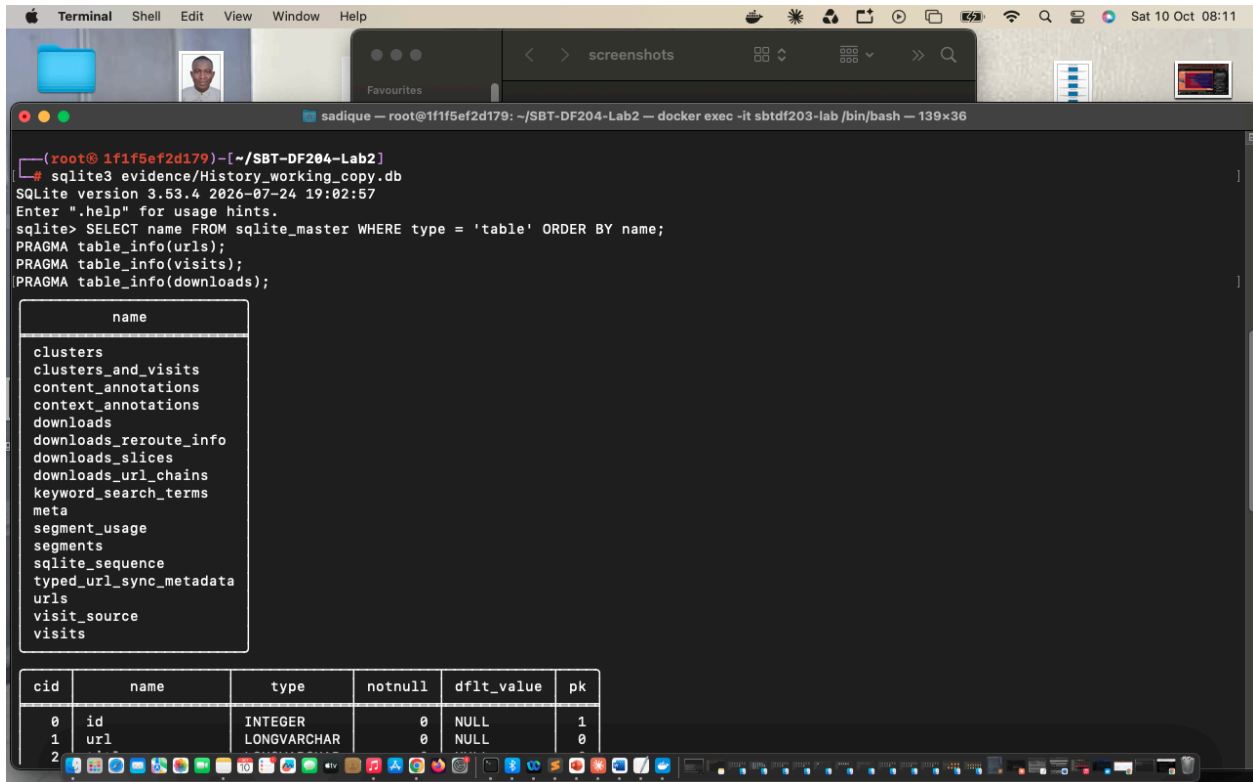
sadique@Abubakari-Sadiques-MacBook-Pro ~ % docker start sbtdf203-lab
sbtdf203-lab
sadique@Abubakari-Sadiques-MacBook-Pro ~ % docker cp /Users/sadique/Desktop/Assignments-ICDFA/forensics/SBT-DF204-Lab2 sbtdf203-lab:/root/
Successfully copied 200kB to sbtdf203-lab:/root/
sadique@Abubakari-Sadiques-MacBook-Pro ~ % docker exec -it sbtdf203-lab /bin/bash
(root@ 1f1f5ef2d179)-[/]
# cd /root/SBT-DF204-Lab2
(root@ 1f1f5ef2d179)-[~/SBT-DF204-Lab2]
# cp evidence/History evidence/History_working_copy.db
(root@ 1f1f5ef2d179)-[~/SBT-DF204-Lab2]
# chmod 444 evidence/History
(root@ 1f1f5ef2d179)-[~/SBT-DF204-Lab2]
#
```

```
Terminal Shell Edit View Window Help
sadique — root@1f1f5ef2d179: ~/SBT-DF204-Lab2 — docker exec -it sbtdf203-lab /bin/bash — 139x36

sadique@Abubakari-Sadiques-MacBook-Pro ~ % docker start sbtdf203-lab
sbtdf203-lab
sadique@Abubakari-Sadiques-MacBook-Pro ~ % docker cp /Users/sadique/Desktop/Assignments-ICDFA/forensics/SBT-DF204-Lab2 sbtdf203-lab:/root/
Successfully copied 200kB to sbtdf203-lab:/root/
sadique@Abubakari-Sadiques-MacBook-Pro ~ % docker exec -it sbtdf203-lab /bin/bash
(root@ 1f1f5ef2d179)-[/]
# cd /root/SBT-DF204-Lab2
(root@ 1f1f5ef2d179)-[~/SBT-DF204-Lab2]
# cp evidence/History evidence/History_working_copy.db
(root@ 1f1f5ef2d179)-[~/SBT-DF204-Lab2]
# chmod 444 evidence/History
(root@ 1f1f5ef2d179)-[~/SBT-DF204-Lab2]
# ls -la evidence/
total 392
drwxr-xr-x 2 501 dialout 4096 Oct 10 07:54 .
drwxr-xr-x 4 501 dialout 4096 Oct 9 22:25 ..
-r--r--r-- 1 501 dialout 196608 Oct 10 07:45 History
-rw-r--r-- 1 root root 196608 Oct 10 07:54 History_working_copy.db
(root@ 1f1f5ef2d179)-[~/SBT-DF204-Lab2]
# sha256sum evidence/History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 evidence/History
(root@ 1f1f5ef2d179)-[~/SBT-DF204-Lab2]
# ls -l evidence/History
-r--r--r-- 1 501 dialout 196608 Oct 10 07:45 evidence/History
(root@ 1f1f5ef2d179)-[~/SBT-DF204-Lab2]
#
```

3. Database Schema Analysis

The SQLite database structure of the Chrome History file was inspected using sqlite3 PRAGMA queries to identify schema layouts and foreign key relationships.



```
(root@1f1f5ef2d179) ~/SBT-DF204-Lab2
# sqlite3 evidence/History_working_copy.db
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> SELECT name FROM sqlite_master WHERE type = 'table' ORDER BY name;
PRAGMA table_info(urls);
PRAGMA table_info(visits);
PRAGMA table_info(downloads);
```

	name
	clusters
	clusters_and_visits
	content_annotations
	context_annotations
	downloads
	downloads_reroute_info
	downloads_slices
	downloads_url_chains
	keyword_search_terms
	meta
	segment_usage
	segments
	sqlite_sequence
	typed_url_sync_metadata
	urls
	visit_source
	visits

cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	0	NULL	1
1	url	LONGVARCHAR	0	NULL	0
2					

cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	0	NULL	1
1	url	INTEGER	1	NULL	0
2	visit_time	INTEGER	1	NULL	0
3	from_visit	INTEGER	0	NULL	0
4	transition	INTEGER	1	'0'	0
5	segment_id	INTEGER	0	NULL	0
6	visit_duration	INTEGER	1	'0'	0
7	incremented_omnibox_typed_score	BOOLEAN	1	FALSE	0
8	opener_visit	INTEGER	0	NULL	0

	cid	name	type	notnull	dflt_value	pk
	0	id	INTEGER	0	NULL	1
1		guid	VARCHAR	1	NULL	0
2		current_path	LONGVARCHAR	1	NULL	0
3		target_path	LONGVARCHAR	1	NULL	0
4		start_time	INTEGER	1	NULL	0
5		received_bytes	INTEGER	1	NULL	0
6		total_bytes	INTEGER	1	NULL	0
7		state	INTEGER	1	NULL	0
8		danger_type	INTEGER	1	NULL	0
9		interrupt_reason	INTEGER	1	NULL	0
10		hash	BLOB	1	NULL	0
11		end_time	INTEGER	1	NULL	0
12		opened	INTEGER	1	NULL	0

visit_id	url_id	visit_utc	url	title	base_transition
1	1	2022-04-19 13:54:18	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i40212j0i51213j0i131i43312j0i512.2595j0j15&...	craigslist - Google Search	5
2	1	2022-04-19 13:54:19	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i40212j0i51213j0i131i43312j0i512.2595j0j15&...	craigslist - Google Search	0
3	2	2022-04-19 13:54:19	https://www.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	0
4	3	2022-04-19 13:54:19	https://geo.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	0
5	4	2022-04-19 13:54:19	https://baltimore.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	0

Core Tables Utilized:

- **urls Table:** Contains primary indexed web addresses, page titles, total visit counts (visit_count), and omnibox typed counts (typed_count).
- **visits Table:** Tracks discrete visit instances, WebKit microsecond timestamps (visit_time), preceding transition links (from_visit), and navigation bitmasks (transition).
- **downloads Table:** Records local file downloading activity, target disk paths, total byte size, and originating tab URL attributes.

4. Reconstructed UTC Event Timeline

Chrome records temporal metrics using WebKit microsecond timestamps (microseconds elapsed since January 1, 1601 00:00:00 UTC). Timestamps were converted into standard ISO-8601 UTC representations using the following transformation:

UTC Time = datetime((visit_time / 1000000) + strftime('%s', '1601-01-01'), 'unixepoch')

The screenshot shows a macOS desktop with a terminal window open. The terminal is running a SQLite query on a database named 'evidence/History_working_copy.db'. The query selects visit_id, visit_time, url, and title from the 'visits' table, joined with the 'urls' table on u.url = u.url. The results are ordered by visit_time. The output shows five rows of data, all with a visit_time of 2022-04-19 13:54:18. The URLs are all from google.com, and the titles are all 'craigslist - Google Search'.

visit_id	visit_time	url	title
1	2022-04-19 13:54:18	https://www.google.com/search?q=craigslist&rlz=1C10W0R-enUS980US989&oeq=craigslist&ags=chrome..69167j46i131i199i433i465...	craigslist - Google Search
2	2022-04-19 13:54:18	https://www.google.com/search?q=craigslist&rlz=1C10W0R-enUS980US989&oeq=craigslist&ags=chrome..69167j46i131i199i433i465...	craigslist - Google Search
3	2022-04-19 13:54:18	https://www.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events
4	2022-04-19 13:54:18	https://geo.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events
5	2022-04-19 13:54:18	https://baltimore.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events

Reconstructed Chronological Sequence of Events

Visit ID	Timestamp (UTC)	Target URL / Resource	Resource Title / Narrative	Transition Type
1	2022-04-19 13:54:18	google.com/search?q=craigslist	Google Search: craigslist	Omnibox Search (5)
3	2022-04-19 13:54:19	baltimore.craigslist.org/	Craigslist Baltimore Homepage	Link Navigation (0)
11	2022-04-19 13:55:05	post.craigslist.org/k/.../choose	Ad Creation: Category Selection	Link Navigation (0)
14	2022-04-19 13:55:40	post.craigslist.org/k/.../edit	Ad Creation: Form Entry	Form Submit (7)
16	2022-04-19 13:57:01	post.craigslist.org/k/.../geoverify	Location Verification	Form Submit (7)
17	2022-04-19 13:57:22	post.craigslist.org/k/.../editimage	Image Upload / Processing	Form Submit (7)
18	2022-04-19 13:58:12	post.craigslist.org/k/.../preview	Listing Preview Stage	Form Submit (7)
20	2022-04-19 13:58:27	baltimore.craigslist.org/hab/d/...	Live Ad: "cheaper than	Link Navigation (0)

Visit ID	Timestamp (UTC)	Target URL / Resource	Resource Title / Narrative	Transition Type
		/7473121658.html	Rx supplements"	
24	2022-04-19 13:58:40	mail.google.com/mail/u/0/	Gmail Session (unsub.fscs@gmail.com)	Link Navigation (0)
53	2022-04-19 14:59:01	blockchain.com/btc/tx/517b2156914...	BTC Tx Lookup (517b215691494433...)	Link Navigation (0)
56	2022-04-19 14:59:12	blockchain.com/btc/address/38RcsURW...	BTC Address Lookup (38RcsURWCDm...)	Link Navigation (0)
DL 1	2022-04-19 14:56:58	imgur.com/dTgrkP7	Downloaded proof_of_payment.png	File Download

5. Navigation Transition Analysis & User Intent

Parsing the transition bitmask (transition & 0xFF) establishes the exact mechanism by which pages were accessed, providing clear insight into user intent:

- **Transition 5 (GENERATED / OMNIBOX):** Indicates deliberate user input into the browser address bar to execute external search queries.
- **Transition 0 (LINK):** Demonstrates sequential navigation resulting from direct user

interaction with hyperlinks.

- **Transition 7 (FORM_SUBMIT):** Proves active manual entry and submission of structured text forms, validating deliberate user progression through the multi-stage Craigslist ad creation wizard.

6. Detailed Investigative Findings

6.1 Illicit Advertisement Artifacts

The suspect navigated through the multi-stage posting wizard on Craigslist Baltimore to post an unauthorized advertisement offering pharmaceutical alternatives.

The screenshot shows a macOS desktop with a terminal window open. The terminal displays a series of blockchain transactions and addresses, along with a sqlite query and its results. The desktop background features various folders and files, including 'Academics', 'projects', 'MY FSSSS', 'resources', and 'AI_Moderation'. The terminal window is titled 'Terminal' and shows the following content:

```
sadique - root@1f1f5ef2d179: ~/SBT-DF204-Lab2 - docker exec -it sbtdf203-lab /bin/bash - 146x47
```

id	timestamp	url	transaction
55	2022-04-19 14:59:06	https://www.blockchain.com/search?search=517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 Blockchain Explorer
56	2022-04-19 14:59:06	https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 Blockchain Explorer
57	2022-04-19 14:59:06	https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 Blockchain Explorer
58	2022-04-19 14:59:21	https://www.blockchain.com/btc/address/38RcsURwCmCbYocmsZCnGz1FtD7x477mt	Address: 38RcsURwCmCbYocmsZCnGz1FtD7x477mt Blockchain Explorer
72	2022-04-19 15:02:32	https://mail.google.com/mail/u/0/#inbox/WhetKXXDXdfcqvtrjhdkMBqVNFnetCnJjRvqlXzZnKFbrkGDZwdVPkdDvjBPxRxDLqKDSv7compose=3HrtffdgkFjdsFdvFzNvsxdZrPvcWRNcNnkkcfKzhGgsQ1bhJdvrmBxZntmQDrNtC...	cheaper than Rx supplements - order - unsub.facs@gmail.com - Gmail
73	2022-04-19 15:02:44	https://mail.google.com/mail/u/0/#inbox/WhetKXXDXdfcqvtrjhdkMBqVNFnetCnJjRvqlXzZnKFbrkGDZwdVPkdDvjBPxRxDLqKDSv7compose=3HrtffdgkFjdsFdvFzNvsxdZrPvcWRNcNnkkcfKzhGgsQ1bhJdvrmBxZntmQDrNtC...	cheaper than Rx supplements - order - unsub.facs@gmail.com - Gmail
74	2022-04-19 15:02:56	https://mail.google.com/mail/u/0/#inbox/WhetKXXDXdfcqvtrjhdkMBqVNFnetCnJjRvqlXzZnKFbrkGDZwdVPkdDvjBPxRxDLqKDSv7compose=3HrtffdgkFjdsFdvFzNvsxdZrPvcWRNcNnkkcfKzhGgsQ1bhJdvrmBxZntmQDrNtC...	cheaper than Rx supplements - txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - unsub.facs@gmail.com - Gmail

```
sqlite> SELECT id AS download_id, datetime(start_time/1000000 + strftime('%s', '1601-01-01'), 'unixepoch') AS start_utc, target_path, tab_url, total_bytes FROM downloads;
```

download_id	start_utc	target_path	tab_url	total_bytes
1	2022-04-19 14:56:58	C:\Users\FSCS_User\Desktop\proof_of_payment.png	https://imgur.com/dTgrkP7	33844

```
sqlite>
```

- **Listing Title:** "cheaper than Rx supplements"
- **Craigslist Listing ID:** 7473121658
- **Live Listing URL:**
<https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/7473121658.html>

Creation Workflow Sequence:

choose type → choose category → edit content → geoverify → edit image → preview → publish

6.2 Email Communications & Cryptocurrency Verification

Correlating URL history with download artifacts established a direct link between customer communications, cryptocurrency verification, and local file storage.

- **Email Identity:** The browser session maintained an active session for Gmail account unsub.fscs@gmail.com.

Cryptocurrency Transaction Verification:

- **Bitcoin Transaction Hash (TXID):**
517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5
- **Bitcoin Receiving Address:** 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt
- **Explorer Platform:** The user actively navigated to blockchain.com to confirm transaction status and balance allocations for this wallet address.

Downloaded Payment Proof:

- **Filename:** proof_of_payment.png
- **Local Storage Path:** C:\Users\FSCS_User\Desktop\proof_of_payment.png
- **Source Hosting URL:** <https://imgur.com/dTgrkP7>
- **File Size:** 33,844 bytes
- **Download Timestamp:** 2022-04-19 14:56:58 UTC

7. Forensic Evaluation & Limitations

- **Identity Attribution:** Browser history artifacts confirm system user account FSCS_User and email handle unsub.fscs@gmail.com performed these actions. However, physical attribution to a specific individual requires supporting correlation with host system security logs, DHCP/IP assignments, and network telemetry.
- **Message Body Data:** While URL strings establish interactions with specific email threads and external image links (Imgur), full email body contents, and message text are not

preserved inside Chrome's History database file and must be recovered from volatile memory, browser cache, or network traffic captures.

8. Conclusion

The forensic analysis of the Chrome History database provides a complete, chronologically sound chain of evidence. The extracted artifacts directly link the user account on April 19, 2022, to the creation and publication of Craigslist ad #7473121658, active correspondence via unsub.fscs@gmail.com, verification of Bitcoin transaction 517b2156... on Blockchain Explorer, and the download of transaction receipt proof_of_payment.png to the local desktop environment.